RISK REGISTER

Customer Churn Prediction Using Machine Learning

Project: Customer Churn Prediction
Domain: Artificial Intelligence and Machine Learning
Purpose: To identify potential risks in developing and using a customer churn prediction model and suggest suitable mitigation strategies.

1. Introduction

A risk register is a document that identifies possible problems that may affect a project's success. In this project, Machine Learning is used to predict whether a customer may stop using a company's services. Risks related to data quality, privacy, fairness, and model performance must be considered.

2. Risk Identification and Mitigation

No.	Risk	Impact	Mitigation Strategy
1	Missing Data	Missing values may affect model performance.	Check missing values and use suitable imputation methods.
2	Incorrect Data	Incorrect records may produce misleading predictions.	Validate data types, values, and consistency.
3	Data Privacy	Customer information may be exposed.	Protect sensitive data and avoid publishing private information.
4	Biased Data	The model may produce unfair predictions for some groups.	Check data representation and compare performance across appropriate groups where possible.
5	Class Imbalance	The model may perform poorly when identifying churn cases.	Examine the target distribution and evaluate precision, recall, and F1-score.
6	Overfitting	The model may perform well on training data but poorly on new data.	Use separate training and testing data and evaluate generalization carefully.
7	Data Leakage	Information from the test data may influence training and produce misleading results.	Split the data before fitting preprocessing steps and the model.
8	False Predictions	Customers may be incorrectly classified as likely to churn or not churn.	Evaluate model performance and require human review before important decisions.
9	Model Performance Changes	Changes in customer behaviour may reduce prediction reliability.	Monitor performance and retrain the model when appropriate.

10	Over-reliance on AI	Decisions based only on predictions may be unfair or inappropriate.	Use predictions as decision-support information and maintain human oversight.

3. Risk Priority

High Priority

- Data privacy violations.
- Biased or unfair predictions.
- Data leakage.
- Decisions made solely on automated predictions.

Medium Priority

- Missing or incorrect data.
- Class imbalance.
- Overfitting.
- Changes in customer behaviour.

These priorities are preliminary and should be reviewed as the project develops.

4. Risk Monitoring Plan

The following actions will help manage risks throughout the project:

1. Inspect the dataset for missing values, duplicates, and inconsistencies.
2. Check the distribution of churn and non-churn records.
3. Evaluate the model using appropriate performance metrics.
4. Document the model's limitations and assumptions.
5. Protect customer information when storing and sharing files.
6. Review predictions before using them for business decisions.

5. Project Limitations

The supplied dataset contains only 12 customer records, including 5 churn cases and 7 non-churn cases. This is too small to establish reliable real-world performance. Evaluation results may vary considerably, and further testing on a larger, representative dataset is needed.

6. Conclusion

The risk register helps identify and reduce potential problems in the customer churn prediction project. Responsible data handling, fairness, data quality, reliable evaluation, and human oversight are essential for developing a trustworthy Machine Learning solution.